

Polynomial Voxels on an Octree

Placing the Cells where the Medium Needs them, with the Error Known before the Solve

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Abstract

A volume-integral model of seismic waves in a realistically complicated medium cannot afford a uniform grid: real media vary on localised scales, and a uniform grid spends most of its cells where nothing varies. Adaptive trees of polynomial cells, with fast solvers, are established for scalar scattering problems; they are refined to a tolerance, and what the tolerance means for the error of the scattered field is not stated. We ask whether, for elastic waves, cubic voxels on an octree can be placed where the medium needs them with the error of the scattering calculation known before it is run. Each leaf of the tree carries the medium's contrast to polynomial degree r and the wavefield to degree p , and couples to every other leaf through blocks that follow exactly from those between equal cells. We find that the error of the scattered far field is the sum of two terms, both available before the solve. The first is the error of the term of first order in the contrast, which needs no solve: each leaf holds the projection of the incident wave, and for a leaf of uniform contrast its error is a closed-form factor in spherical Bessel functions, with leading term $-(\mathbf{k}_{\text{in}} \cdot \mathbf{k}_{\text{out}}) d^2/12$ for a constant field in a cell of side d . The second is the nonlinear fraction of the response times the relative projection error of the medium. On a sphere with a thin graded shell, against its exact solution, the second term has one constant (0.015) for constant and for linear cells, on uniform grids and on trees, where the raw ratio of error to projection error varies by a factor of two. Refining by both terms, a tree of linear cells reaches the accuracy of a uniform grid with a quarter of the cells on a body whose variation fills 3% of its volume. Which basis suits the medium depends on the medium. Where it varies smoothly the two degrees must be matched, and a constant contrast with a richer field is worse than a constant field. Where it is piecewise constant on the leaves, a constant contrast is exact, and a linear field on the tree has 65 times less error than the uniform grid of constant cells with the same number of unknowns. This draft records what is established and what is not.

1 Introduction

The problem this paper serves. The aim of this series is an efficient model of the seismic wavefield in a medium of realistic complication. Its method is the volume integral equation of elastic scattering [1], discretised into cubic voxels that interact through the Green's tensor of a background medium, the multiple-scattering description of Foldy [2] and Lax [3]. The first

paper of the series [4] determined what a voxel must carry. A voxel represents two functions, the medium within it and the wavefield within it, each in a basis of its own. With the wavefield held to Legendre degree p and the contrast of the medium to degree r , the scheme converges at order $\min(2p + 2, 2r + 2)$ in the cell size, and the error that remains belongs to the single voxel: at second order in the contrast it is the *relative projection error* of the medium, the fraction of the squared norm of its profile that polynomials of degree $\min(p, r)$ on the cells cannot represent. The companion paper [5] builds those voxels in three dimensions.

The question. Those results are for uniform grids. A real medium is not uniform in its difficulty: its variation is concentrated at interfaces, inclusions and thin zones, and a uniform grid fine enough for those is far finer than the rest of the medium needs. The natural cells are then the leaves of an octree, cubes whose sizes differ by powers of two. This paper asks two things. Can the cells be placed where the medium needs them? And can the error of the calculation be known before it is run, from the medium, the tree and the wave alone? The second matters as much as the first. A tree is built by a rule, the rule needs an indicator of where the error is, and an indicator that needs the solution is too late.

What exists. The tree is not new. For the scalar Lippmann–Schwinger equation in two dimensions, Ambikasaran et al. [6] discretise on an adaptive quad tree with a polynomial on every leaf, splitting a leaf until it holds a set number of points per wavelength and resolves the contrast to a tolerance, and Gopal and Martinsson [7] accelerate the direct solve. In three dimensions Malhotra and Biros [8, 9] represent volume sources by polynomials of bounded total degree on the leaves of an adaptive octree, refined where the highest coefficients exceed a tolerance, and evaluate their potentials with a kernel-independent fast multipole method [10] for the Poisson, Stokes and low-frequency Helmholtz problems, at up to 10^{10} unknowns. Polynomials on the cells of a dyadic tree are the multiwavelets of Alpert [11], in which integral operators are sparse [12]. In electromagnetics, large volume elements carry fields of high polynomial order [13], with the material represented to an order of its own [14], and voxels with a linear field are solved with fast Fourier transforms [15]. Where error is estimated for such methods it is estimated *a posteriori*, from a computed solution that is then refined and solved again [16]. For elastic waves the volume integral equation is solved on uniform grids at low order [17], and fast multipole methods exist for boundary integral equations [18, 19].

What this paper adds. Not the tree, the adaptivity or a fast solver. It adds the statement of the error before the solve, for the elastic problem: the two terms of §3, the closed-form factor of a leaf, and the rule for choosing the basis of the medium. The law is a sharpened instance of a standard fact, that the error of a Galerkin solution of a second-kind equation is governed by the best approximation of the solution; here it is an equality with a known constant, for the part of the field that copies the medium.

The answer in brief. The error of the scattered field separates by its order in the contrast into two terms, and both are available before the solve (§3). The term of first order is computed

in one pass over the leaves, and for a leaf of uniform contrast it is a closed form. The rest is the nonlinear fraction of the response times the relative projection error of the medium, with one constant for every grid and tree tried. A refinement rule built on the two terms produces trees that a rule built on the medium alone cannot (§4). And the choice of basis for the medium follows from the medium (§5): a constant in each leaf is the right basis where the medium is piecewise constant on the leaves, and the wrong one where it varies smoothly.

2 Setting

Leaves. The body is covered by cubes, the leaves of an octree: leaf ℓ has centre $\mathbf{x}_\ell$ and half-width h_ℓ (side $d_\ell = 2h_\ell$), and the half-widths are in ratios that are powers of two. With $\boldsymbol{\xi} = (\mathbf{x} - \mathbf{x}_\ell)/h_\ell$ the local coordinate, the leaf carries the nine-component state (displacement and strain) in the Legendre products $Q_a(\boldsymbol{\xi}) = P_{a_1}(\xi_1)P_{a_2}(\xi_2)P_{a_3}(\xi_3)$ of total degree $|a| = a_1 + a_2 + a_3 \leq p_\ell$, which is 1, 4 or 10 functions for $p_\ell = 0, 1$ or 2 , and the contrast in the same products to degree r_ℓ . The two degrees may differ from leaf to leaf. A leaf with $p = r = 0$ is a Haar cell: a constant medium and a constant field.

The system. Tested with the same functions, the volume integral equation becomes

$$M_\ell \psi_\ell - \sum_m \mathbf{K}(\ell, m) E_m \psi_m = \langle Q, \psi^0 \rangle_\ell, \quad (1)$$

with ψ_ℓ the coefficients of the state in leaf ℓ , M_ℓ the Gram matrix of its functions, E_m the projected contrast of leaf m acting on its field, ψ^0 the incident wave and $\mathbf{K}(\ell, m)$ the block that couples the two leaves: the Green's tensor integrated against the functions of both [5].

Blocks between leaves of different sizes. A polynomial on a cube is a polynomial of the same degree on each of the cube's descendants, so the block between a large leaf and a small one is an exact, finite sum of blocks between equal cells: the large leaf is replaced by its descendants at the size of the small one, with its polynomials re-expanded. No integral is needed beyond those between equal cells. This is the two-scale relation of the multiwavelet basis [11]. Against six-dimensional quadrature these blocks agree to 10^{-9} for leaves one and two levels apart, and on a tree whose leaves are all equal the solver reproduces the uniform solver to 10^{-11} .

Symmetry. Two pairs of leaves whose relative positions are images of one another under a reflection or a permutation of the axes have blocks related by the representation of that operation on the polynomials and on the nine components, whatever the two sizes, because the operation acts on both cubes alike. One block is therefore computed for each orbit of the cube's group of 48 operations. On a tree of 344 leaves of three sizes this reduced the assembly from 125 s to 65 s with an unchanged solution, and made trees with four sizes practicable (§6).

3 The error in two terms

Write the scattered far field as a series in the contrast: $u = T_1 + T_2 + \dots$, with T_1 the term of first order (the Born term) and the rest the nonlinear part of the response. The error of the scheme separates the same way.

3.1 The first-order error needs no solve

At first order in the contrast Eq. (1) is $M_\ell \psi_\ell = \langle Q, \psi^0 \rangle_\ell$: each leaf holds the L^2 projection of the incident wave onto its field functions. The scheme's Born term is therefore known from one pass over the leaves, with no system to solve, and its difference from the Born term of a finer representation (each leaf's eight children with quadratic cells) is the first-order error of the tree, leaf by leaf. On a tree of two leaf sizes this projected Born term equals the weak-contrast limit of the full solver to 10^{-5} of its size.

3.2 The wave factor of a leaf

For a leaf whose contrast is uniform the first-order error is a closed form. The leaf represents the incident wave $e^{i\mathbf{k}_{\text{in}} \cdot \mathbf{x}}$ by its projection, and it radiates towards the observer through the moments of $e^{-i\mathbf{k}_{\text{out}} \cdot \mathbf{x}}$, with $\mathbf{k}_{\text{out}}$ the P or the S wavevector in the direction of observation. Since $\int_{-1}^1 P_n(t) e^{iq t} dt = 2i^n j_n(q)$, with j_n the spherical Bessel function, the leaf's Born far field is its exact Born far field times

$$1 + E_p = \frac{\sum_{|a| \leq p} \prod_{i=1}^3 (2a_i + 1) j_{a_i}(k_{\text{in},i} h) j_{a_i}(k_{\text{out},i} h)}{\prod_{i=1}^3 j_0((k_{\text{in},i} - k_{\text{out},i}) h)}. \quad (2)$$

The sum over all degrees is the denominator (the addition theorem), so $E_p \rightarrow 0$ as p grows. Its leading term is

$$E_p = -h^{2p+2} \sum_{|a|=p+1} \prod_{i=1}^3 \frac{(2a_i + 1) (k_{\text{in},i} k_{\text{out},i})^{a_i}}{[(2a_i + 1)!!]^2} + O(h^{2p+4}), \quad E_0 = -\frac{(\mathbf{k}_{\text{in}} \cdot \mathbf{k}_{\text{out}}) d^2}{12} + \dots \quad (3)$$

For a constant field the error of a leaf therefore depends on the angle of scattering, vanishes at right angles to the incident wave, and differs between the P and the S far field. Along one axis, with $\mathbf{k}_{\text{out}} = \pm \mathbf{k}_{\text{in}}$ of magnitude k , Eq. (3) is $-(\pm 1)^{p+1} c_p (kd)^{2p+2}$ with $c_p = (2p+3)/(4^{p+1}[(2p+3)!!]^2)$: the constants $c_0 = 1/12$, $c_1 = 1/720$ and $c_2 = 1/100800$ of the layer [4]. The layer's closed-form error is the one-dimensional case of the leaf's.

Checks. Equation (2) is derived symbolically, built a second way without Bessel functions (project, multiply, integrate over the cube) and found identical for $p = 0, 1, 2$; Eq. (3) and its one-dimensional reduction are verified symbolically; and the implementation used below agrees

with 30-digit values of the factor to 10^{-13} in 36 cases. Against the solver, on a homogeneous cube (which every grid of cubes represents exactly) at $k_s A = 1$ for half-side A : the Born far field divided by $1 + E_p$ agrees across $p = 0, 1, 2$ and two grids to 9×10^{-7} , where the uncorrected fields differ by up to 14%; and the Born far field of a tree of 15 leaves of mixed sizes and degrees equals the sum over its leaves of exact contribution times factor to 2×10^{-9} .

3.3 The rest is the projection error of the medium

Let $\mathcal{E}_r$ be the relative projection error of the medium on the tree,

$$\mathcal{E}_r = \frac{\sum_\ell \int_\ell (f - \Pi_r f)^2 dV}{\int f^2 dV}, \quad (4)$$

with $f(\mathbf{x})$ the profile of the contrast and Π_r the L^2 projection onto the polynomials of degree at most r on each leaf. Subtract from the error of the far field the first-order error of §3.1. What is left, divided by $\mathcal{E}_r$, is one constant (Table 1). The test body is a sphere of radius a with a uniform core of radius $0.75a$ and a shell across which the contrast falls smoothly to zero, at $k_s a = 0.5$, whose exact solution is known; the error is the largest difference of the far field from the exact one over nine scattering angles, relative to its peak. For constant cells the ratio of error

Table 1. The error of the far field in two terms, on the sphere with a graded shell. The rest is the error less the first-order error, which is computed without a solve. $\mathcal{E}$ is the relative projection error of Eq. (4) at the degree of the cells. The nonlinear fraction of the exact response is 0.0151.

cells	grid	leaves	error	error / $\mathcal{E}$	rest / $\mathcal{E}$
constant ($p = r = 0$)	uniform	64	8.17×10^{-3}	0.0309	0.0149
	uniform	184	3.18×10^{-3}	0.0235	0.0151
	uniform	408	1.61×10^{-3}	0.0196	0.0154
	uniform	1256	6.77×10^{-4}	0.0166	0.0158
	tree	232	3.29×10^{-3}	0.0321	0.0152
	tree	856	9.28×10^{-4}	0.0210	0.0152
	tree	1360	6.03×10^{-4}	0.0210	0.0151
	tree	1696	5.75×10^{-4}	0.0222	0.0151
linear ($p = r = 1$)	uniform	64	8.99×10^{-4}	0.0147	0.0148
	uniform	184	4.62×10^{-4}	0.0146	0.0146
	uniform	408	2.20×10^{-4}	0.0146	0.0146
	tree	184	5.16×10^{-4}	0.0142	0.0148
	tree	352	2.20×10^{-4}	0.0146	0.0145

to projection error varies by a factor of 1.93 across the grids; the rest over the projection error varies by 1.06, and it is 0.0151 to 0.0152 on all four trees. For linear cells the first-order error is small and both ratios are near 0.0146. The constant is the nonlinear fraction of the exact response, $|u - T_1|/|u| = 0.0151$. This is the law of the layer [4] in three dimensions: the error of the nonlinear part is that part times the relative projection error of the medium. It holds for constant cells as it does for linear ones, once the first-order error is accounted for; without that, constant cells appear not to follow it.

4 Refinement by the two terms

The rule. Each leaf has two indicators, both computed before any solve. The *medium* indicator is the nonlinear fraction ν times the leaf's share of the projection error, $\nu \int_{\ell} (f - \Pi_r f)^2 dV / \int f^2 dV$. The *wave* indicator is the leaf's own first-order error (§3.1) relative to the peak of the Born far field. A leaf is split into its eight children while the sum of the two exceeds a tolerance. The nonlinear fraction is estimated from one solve on a coarse tree that resolves the medium, as $|u - T_1|/|u|$ of that tree. The predicted error of a tree is the first-order error of the whole tree, summed over the leaves as complex amplitudes, plus $\nu \mathcal{E}_r$. It uses no reference solution.

A localised feature. The test body is a sphere of radius a whose contrast is a weak halo, falling smoothly from 5% of the central value to zero across the whole radius, plus a compact feature in which it falls from the central value to the halo between $0.1a$ and $0.3a$: under 3% of the volume. Its exact solution is known. At $k_S a = 1$, with leaves no smaller than $a/32$, the errors are those of Table 2. The rule that uses the medium alone refines a leaf while its share of the projection error exceeds a tolerance.

Table 2. The localised feature: leaves, the error predicted without a reference, and the error against the exact solution. The nonlinear fraction estimated from a coarse tree is 0.0051 (constant cells) and 0.0059 (linear cells); the exact value is 0.0059.

cells	grid	leaves	predicted	error
constant	uniform	408	1.11×10^{-2}	7.15×10^{-3}
	uniform	1256	4.36×10^{-3}	2.92×10^{-3}
	medium alone	344	1.60×10^{-2}	1.55×10^{-2}
	medium alone	512	1.57×10^{-2}	1.53×10^{-2}
	medium alone	736	5.92×10^{-3}	5.49×10^{-3}
	two terms	260	8.18×10^{-3}	7.18×10^{-3}
	two terms	372	6.82×10^{-3}	5.38×10^{-3}
	two terms	752	4.48×10^{-3}	3.78×10^{-3}
linear	two terms	1780	2.18×10^{-3}	1.75×10^{-3}
	uniform	184	2.83×10^{-3}	2.73×10^{-3}
	uniform	408	8.27×10^{-4}	6.77×10^{-4}
	medium alone	176	2.43×10^{-4}	2.47×10^{-4}
	medium alone	344	1.95×10^{-4}	1.96×10^{-4}
	two terms	260	1.72×10^{-4}	1.68×10^{-4}
	two terms	336	1.56×10^{-4}	1.66×10^{-4}
	two terms	452	1.08×10^{-4}	1.14×10^{-4}

Constant cells. The rule that uses the medium alone fails: with 512 leaves its error is twice that of the uniform grid of 408 cells. It refines the feature and leaves the halo coarse, and the halo is where the first-order error is, in leaves whose medium is too smooth for the rule to see. The rule with two terms sees it. Its tree of 260 leaves has the error of the uniform grid of 408, and against the trend of the uniform grids its trees have the same error with about 35% fewer cells at the coarse end and an error 15 to 20% lower at the fine end. The saving is modest, and

the reason is the cell: a constant field has a second-order error in every leaf, so the halo needs cells however smooth its medium is.

Linear cells. Here the tree pays. The tree of 176 leaves has eleven times less error than the uniform grid of 184 cells. By the trend of the two uniform grids its error would need about 730 uniform cells, four times as many, and that of the tree of 452 leaves about 1130; these uniform equivalents are extrapolated, not computed. The fourth-order error of a linear field lets the halo stay coarse, and the tree then follows the medium. The two-term rule improves on the medium alone by about 15% at about 340 leaves.

The prediction. The predicted error is above the true error on all eleven grids of constant cells computed for this body (nine of them in Table 2), by 3% to 55%. For linear cells the ratio of predicted to true is 0.94 to 1.22 on nine grids: close, but not a bound.

5 Which basis for the medium

A smooth medium: the degrees must be matched. On the body of §4, hold the contrast constant in every leaf and raise the field to linear. The error rises, at four times the unknowns: from 7.15×10^{-3} to 8.31×10^{-3} on the uniform grid of 408 cells, and from 7.18×10^{-3} to 8.32×10^{-3} on the tree of 260 leaves. The prediction, made without a solve, says so beforehand (1.11×10^{-2} against 1.34×10^{-2} on the uniform grid). On a smoothly varying medium the first-order error of a leaf with a constant contrast is not the wave's: it is the error of holding the contrast constant, which a richer field cannot repair, and with a constant field the wave's error partly cancels it. This is the rule of Nestor [4] that the two bases must be matched.

A medium that is piecewise constant on the leaves. The pairing of a constant medium with a polynomial field belongs to the medium that is constant in each leaf. The test body is a cube of half-side B with a uniform contrast a quarter of the central value, holding at its centre a cube of half-side $B/4$ with the full contrast, at $k_5 B = 2$. A tree of 120 leaves represents it exactly with one constant per leaf ($\mathcal{E}_0 = 0$): 56 leaves of half-width $B/4$ and 64 of half-width $B/8$. The uniform grid that does the same has 512 cells. No exact solution is known for this body; the reference is the tree with quadratic fields, whose own first-order error is predicted to be 2×10^{-7} (Table 3). With the medium exact there is no second term, and the first-order prediction is within 13% of the measured error in all seven cases and within 4% in five. The tree with a linear field has an error of 8.4×10^{-5} with 4320 unknowns; the uniform grid of constant cells has 5.5×10^{-3} with 4608: 65 times more. Raising the field to linear in the large leaves alone lowers the tree's error from 1.5×10^{-2} to 3.2×10^{-3} : a large leaf needs a richer basis for the wavefield, not for the medium. And where the faces of the inner cube fall inside cells, the linear field is worse than the constant one, as on the smooth body; for the scalar problem Anderson et al. [20] likewise find that high order is lost at jumps of the medium unless the grid follows them.

Table 3. A cube within a cube, the contrast constant in every leaf. The first-order error is predicted without a solve; the error is against the tree with quadratic fields. In the last two rows the faces of the inner cube fall inside the cells of the grid.

grid and field	cells	unknowns	predicted	error
tree, constant field	120	1080	1.54×10^{-2}	1.53×10^{-2}
tree, linear in the 56 large leaves	120	2592	3.24×10^{-3}	3.21×10^{-3}
tree, linear field	120	4320	8.07×10^{-5}	8.42×10^{-5}
uniform, constant field	512	4608	5.54×10^{-3}	5.51×10^{-3}
uniform, linear field	512	18432	8.17×10^{-6}	9.25×10^{-6}
uniform, constant field	64	576	1.81×10^{-2}	1.79×10^{-2}
uniform, linear field	64	2304	3.01×10^{-2}	2.92×10^{-2}

The choice. The two cases give the rule. Where the medium is piecewise constant and the tree can follow its interfaces, hold it constant in each leaf and spend the degree on the field. Where it varies smoothly, give the medium and the field the same degree and let the tree follow the medium. In both the error is predicted before the solve.

6 What is established, and what is not

- *Established twice.* The wave factor, Eqs. (2) and (3): a symbolic derivation and an independent numerical implementation, and both against the solver.
- *Established against exact solutions, by one implementation.* The two-term separation (Table 1) and the refinement results (Table 2): one frequency each, two bodies, both spheres with radial profiles.
- *Self-consistent only.* The cube within a cube (Table 3) has no exact solution; its reference is the same scheme with quadratic fields. The field near the edges and corners of the cubes is not smooth, and an independent solution is needed before the figure of 65 is relied on.
- *Not yet derived.* The nonlinear fraction ν is estimated from a coarse solve; it is within 20% of the exact value in the cases tried. The constant of the second term is observed to equal the nonlinear fraction of the response; the layer’s derivation [4] has not been carried to three dimensions.
- *Not a bound.* The predicted error exceeded the true error for constant cells in every case, but fell 6% below it for linear cells.
- *Size.* The solver is dense and stops near 20 000 unknowns: about 2200 constant leaves or 550 linear ones. The saving in cells is therefore shown on small bodies. An equal-cell block costs about 50 ms, and a tree of 344 leaves of four sizes is assembled and solved in about three minutes.
- *Earlier work.* The survey behind §1 found no statement of the two-term law, of the leaf’s factor or of the matched-degree rule, and no elastic volume scheme with polynomial cells.

It rests on some twenty searches and on the opening pages of most of the works cited; the adaptive solvers of Ambikasaran et al. [6] and Malhotra and Biros [9] have yet to be read in full for any estimate that relates their refinement tolerance to the error of the solution.

7 Conclusions

On an octree of polynomial voxels the error of the scattered field is two terms, and both are known before the solve: the first-order error, from one pass over the leaves and in closed form for a leaf of uniform contrast, and the nonlinear fraction of the response times the relative projection error of the medium. The closed form reduces in one dimension to the constants of the layer. A refinement rule built on both terms places cells where a rule built on the medium alone cannot, and with linear cells a tree reaches the accuracy of a uniform grid with a quarter of the cells on a body whose variation is localised. A constant contrast in each leaf is the right basis for a medium that is piecewise constant on the leaves, where a linear field on the tree outperforms the uniform grid of constant cells sixty-five-fold at equal unknowns in a self-consistent test, and the wrong one for a smooth medium, where the two degrees must be matched. What remains before the method models a realistic medium is a solver whose cost does not grow as the square of the number of leaves. Its parts are published: the fast multipole method for volume potentials on an octree [9], and the treatment of the elastic kernels in fast multipole methods for boundary integral equations [18, 19].

The octree, its blocks and its solver are `cubic_scattering/graded_voxel/octree.py`, with tests in `cubic_scattering/tests/test_graded_voxel_octree.py`. The wave factor is derived in `Mathematica/AdaptiveOctree_WaveFactor.wl` and measured by `scripts/measure_octree_wave_term.py`; Table 1 is `scripts/measure_octree_two_term.py`; Table 2 is `scripts/pilot_octree_two_term_refinement.py`; the constant medium with a richer field is `scripts/pilot_octree_hp_refinement.py`; Table 3 is `scripts/pilot_octree_blocky_body.py`. Their results are stored in `LatexPDFs/AdaptiveOctree/figures`.

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